Sequential multivariate change-point detection by betting: algorithms and application to infectious disease monitoring

CANSSI Postdoc Spotlight

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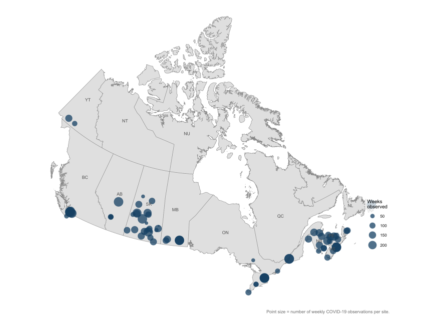
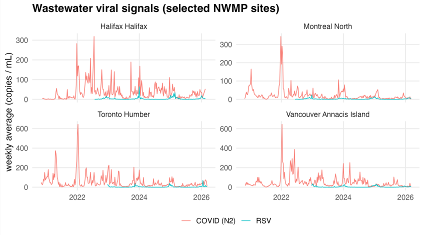
Motivation: online monitoring viral signals in wastewater

Problem: Continuously monitor data streams for "anomalies".





This talk: Construct detectors and apply to viral signals in National Wastewater Monitoring of Pathogens NWMP data.

Roadmap

1. **Part 1 ($\sim$20 min):** Detection-by-betting
    - Goals: validity and efficiency
    - Martingales, testing-by-betting, and the Shiryaev–Roberts procedure
    - Joint detectors via combining
    - Localization via allowances
    - Simulation snapshot
2. **Part 2 ($\sim$10 min):** Wastewater surveillance

Part 1 — Detection-by-betting

The sequential multivariate change-point problem

Continuously observe a stream of K-vectors $(\boldsymbol{X}_t)_{t \in \mathbb{N}}$ such that:

$$(\boldsymbol{X}_t)_{t \leq \nu} \sim P \in \mathcal{P}, \qquad (\boldsymbol{X}_t)_{t > \nu} \sim Q \in \mathcal{Q}.$$

$\nu \in \mathbb{N} \cup \{\infty\}$ is the unknown *change-point*

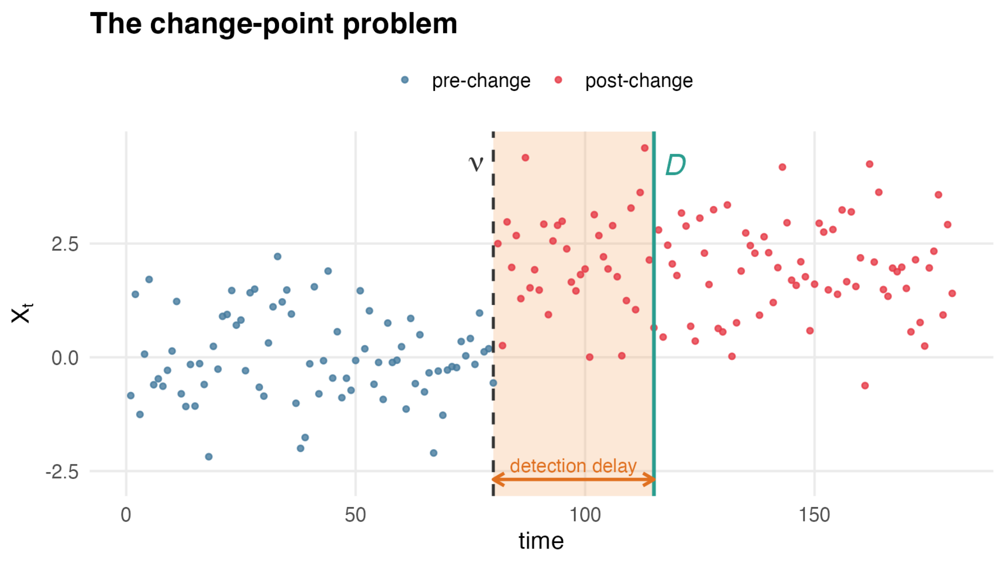
Full change-point model is $(\mathcal{P}, \nu, \mathcal{Q})$

A detector D is a stopping time. We want:

- **Validity:** alarm rate is bounded before change (sequentially, non-asymptotically)
- **Efficiency:** short delay after change.

Conceptual credit: testing-by-betting (Ramdas et al., 2023; Grünwald et al., 2024) + E-detectors (Shin et al., 2024)

Example


The change-point problem
pre-change
post-change
ν
D
2.5
0.0
-2.5
X_t
detection delay
0
50
100
150
time

Applications

- Industrial product quality (Shewart, 1931) and radiolocation (Shiryaev, 1963)
- Linear control systems in aviation and robotics (e.g., Willsky and H. (1976))
- Online supervised learning (Vovk, 2021) and clinical AI (Feng et al., 2022)
- Remote health monitoring (BP, EKG, etc.) (Kishore and Kocher, 2026)
- Anomalies in sensor or computer networks (Xie and Siegmund, 2013)
- Structural change in financial models (Chu et al., 1996; Lyu and Nkurunziza, 2024)

Validity: ARL and PFA

Two error criteria:

$$\mathrm{ARL}(D) := \sup_{P \in \mathcal{P}} \mathbb{E}_P[D]$$

$$\mathrm{PFA}(D) := \sup_{P \in \mathcal{P}} \mathbb{P}_P(D < \infty)$$

D is ARL-*valid at level* α if $\mathrm{ARL}(D) \geq 1/\alpha$

D is PFA-*valid* if $\mathrm{PFA}(D) \leq \alpha$.

Validity is *non-asymptotic* $+$ *time-uniform* (sequential)

This talk: ARL

Efficiency

Conditional average delay: $\mathrm{CAD}(D) := \sup_{\nu \geq 1} \mathbb{E}_{P,\nu,Q}[(D - \nu) \mid D \geq \nu]$

- Averages delay over sample paths where the detector *doesn't* false alarm.
- Define *regret* of D relative to an oracle $D^\star$ that knows (P, Q):

$$\mathrm{regret}(D) := \mathrm{CAD}(D) - \mathrm{CAD}(D^\star).$$

- An *efficient* detector minimizes regret

Goal of this work:

definitely valid, approximately efficient, computationally tractable, multivariate detectors.

Detectors from test supermartingales: Shiryaev–Roberts

Given a *test supermartingale* (M_t) for $\mathcal{P}$ with "increments" $\Lambda_t := M_t/M_{t-1}$ define

$$\boxed{R_t := \Lambda_t(R_{t-1} + 1), \quad R_0 := 0} \qquad \text{and} \qquad \boxed{D_{\text{S-R}} := \inf\{t : R_t > 1/\alpha\}}.$$

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Theorem: $D_{\text{S-R}}$ is ARL-valid for $\mathcal{P}$

Proof sketch: $(R_t - t)$ is a TSM for $\mathcal{P}$, apply optional stopping theorem

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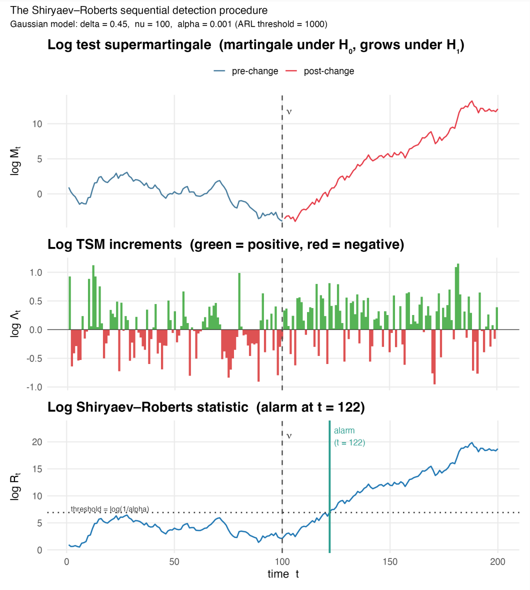
Stats/ML view: TSMs are often (extended) likelihood ratios or conformal P-values

Investment view: At every step t, invest a fresh \$1 to start a new TSM betting against P and compute R_t as the *running sum* of your wealth

Composite pre-change: bet against every $P \in \mathcal{P}$, R_t is the smallest wealth

NB: pre-estimating P by $\widehat{P} \in \mathcal{P}$ is very sensitive to mis-specification (Mei, 2006)

Picture: the Shiryaev–Roberts procedure



Nonparametric likelihood ratio TSM

For a data stream (X_t) bounded below and predictable betting sequence (λ_t):

$$M_t := \prod_{i=1}^{t} [1 + \lambda_i (X_i - \eta_i)]$$

is a TSM when $\sup_{P \in \mathcal{P}} \mathbb{E}_P[X_t | \mathcal{F}_{t-1}] \leq \eta_t$ and $\lambda_t \in [0, 1/\eta]$ for all t.

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Features:

- Finite-sample validity over nonparametric $\mathcal{P}$
- Includes: Binomial, Beta, Poisson, truncated normal, contingency tables, finite-populations, distributions with physical constraints, within-stream dependency

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Strategies:

- Adaptive betting (Waudby-Smith and Ramdas, 2024; Spertus, 2024)
- Mixture betting (Orabona and Jun, 2022; Waudby-Smith et al., 2025; Spertus et al., 2025)
- Handle nuisance parameters by further minimization (Spertus et al., 2024)

Multivariate detectors

Let (M_{tk}) be any marginal TSM for stream k, with increments $\Lambda_{tk} := M_{tk}/M_{(t-1)k}$.

Joint detection: fire if *any* stream changes

- **Combining:** product ($\prod_k M_{tk}$), average ($\sum_k w_k M_{tk}$), universal portfolio
- Product: most powerful, requires cross-stream independence (Vovk and Wang, 2021)
- Universal portfolio: larger than $\max_k M_{tk}$, computationally expensive (Cover, 1991)
- Analogous to intersection testing (meta-analysis)

Localized detection: locate K distinct change-points + control time to first detection

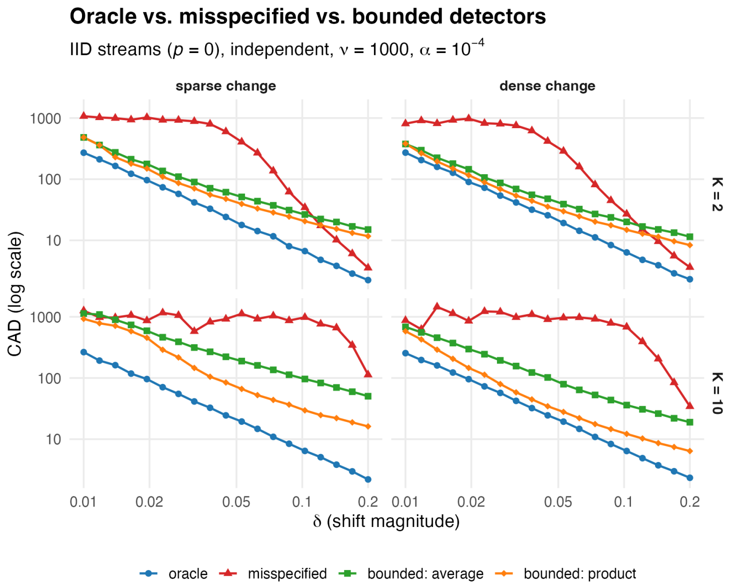
- **Spending allowance**: sum of per-time investment across streams = 1
- Generalizes Bonferonni correction (threshold = K/α)
- Advantages: prioritize streams, adapt to past data, employ side information, dependency graphs, Bayesian models that retain frequentist validity
- Analogous to multiple testing

Simulation setup

K-variate Gaussian VAR model: $\boldsymbol{X}_t = \boldsymbol{\mu} + \sum_{j=1}^{p} \boldsymbol{\Phi}_j(\boldsymbol{X}_{t-j} - \boldsymbol{\mu}) + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$

- $\mathcal{P}$: $\boldsymbol{\mu}_0 := 0.3 \times \mathbf{1}$, $\boldsymbol{\Phi} = \mathbf{0}$, $\boldsymbol{\Sigma} = \mathrm{diag}(0.05^2)$
- $\mathcal{Q}$: $\boldsymbol{\mu}_1 := \boldsymbol{\mu}_0 + \mathbf{1} \times \delta/\sqrt{K}$ (*dense change*) or $\boldsymbol{\mu}_1 := \boldsymbol{\mu}_0 + \boldsymbol{e}_1 \times \delta$ (*sparse change*)
- Four detectors:
  - Oracle TSM (true Gaussian VAR)
  - Bounded TSM + adaptive bets + product combining
  - Bounded TSM + adaptive bets + average combining
  - Mis-specified Gaussian VAR TSM (plug-in $\boldsymbol{\mu}_1 + 0.2 \times \mathbf{1}$)
- Model extensions: streams K, coefficients Φ, covariance $\boldsymbol{\Sigma}$, order p, DGPs
- Detector extensions: universal portfolio, localization, PFA control

Optimal CAD and regret in Gaussian VAR



Part 2 — Wastewater surveillance

Wastewater monitoring as a multi-stream change-point problem

Setting. Canadian NWMP: a graph of wastewater treatment sites producing weekly viral signal time series for COVID, RSV, influenza, . . .

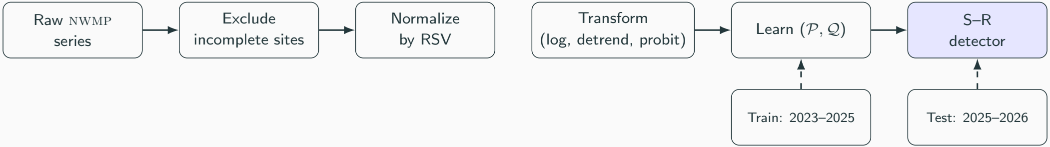
Public-health use case. Past work (Peng et al., Ottawa) links viral signals to leading clinical indicators (hospitalizations 1–3 weeks out).

Detectors: flag sites, triage resources, new strains, model lapses, root cause investigation.

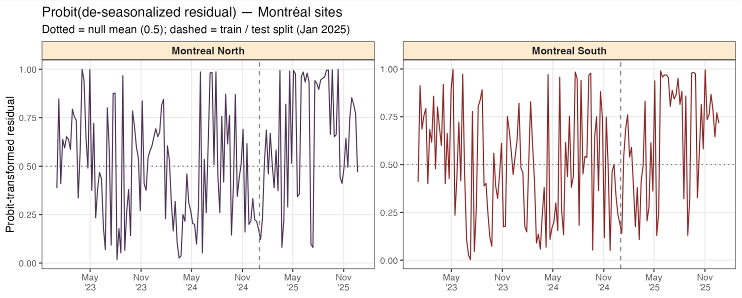
Sources of noise and systematic uncertainty

- **Catchment:** boundaries differ, may not align with clinical catchment areas.
- **Water volume:** open vs. closed sewage systems; weather events; normalization?
- **Sampling design:** location and time; auto vs. grab; compositing; lab protocols.
- **Demography:** secular and seasonal shifts in catchment population
- **Endemicity:** seasonality $\implies$ periodically elevated signals are "in control"
- **Recency:** monitoring starts in 2021 at the earliest sites

Pipeline

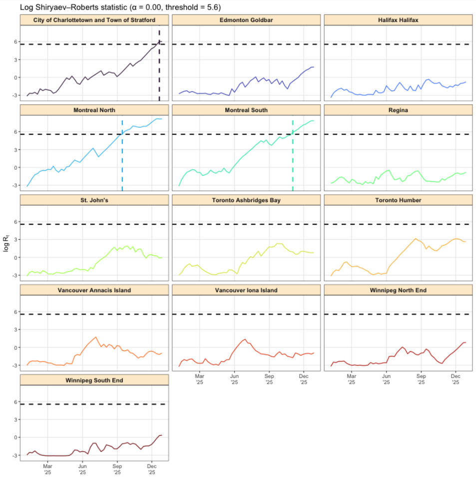


Raw NWMP series
Exclude incomplete sites
Normalize by RSV
Transform (log, detrend, probit)
Learn ($\mathcal{P}$, $\mathcal{Q}$)
S–R detector
Train: 2023–2025
Test: 2025–2026



Probit(de-seasonalized residual) — Montréal sites
Dotted = null mean (0.5); dashed = train / test split (Jan 2025)
Montreal North
Montreal South
Probit-transformed residual
1.00
0.75
0.50
0.25
0.00
May '23
Nov '23
May '24
Nov '24
May '25
Nov '25

Empirical: detector trajectory on 2025–2026 season



log R_t vs. time with threshold $\log(K/\alpha)$ for $\alpha = 1/(52 * 5)$ and $K = 13$.

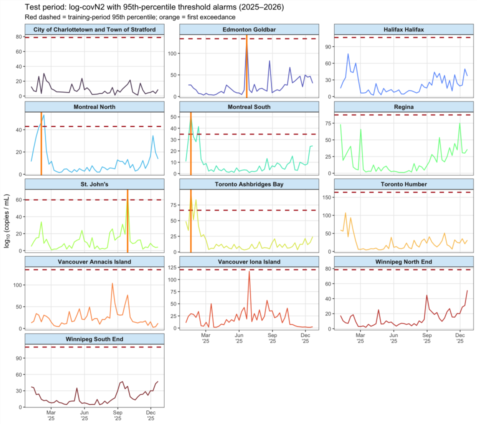
Optimized against Beta(1.5,1) alternative

Montreal alarms around October

Prince Edward Island in December

Detectors fire when ratios are *sustained* above training period levels

Simplicity redux: threshold raw signals?



Alarm times with simple threshold rule

Take some (e.g. 95th) percentile of raw COVID signals on 2023-2025 data

Alarm if threshold is exceeded in future

Edmonton alarms in July, Montreal and Toronto in February, St John's in September

Simple and transparent

Statistical properties are unknown...

But true DGP is unknown right now anyway

Synthesize with betting TSM

Wrap-up

Summary

- Detection-by-betting: Shiryaev–Roberts + TSMs $\Rightarrow \mathrm{ARL}$ control.
- Multivariate combining is meta-analysis: product / average / universal portfolio.
- Localization is multiple testing with a predictable spending allowance.
- Application to wastewater viral monitoring
- R package: `github.com/spertus/multivariate-change-point-detection/`

Extensions + looking ahead

- **Detection-by-betting:** PFA control with spending schedule; efficient computation; graph-adaptive spending; Bayesian spending
- **Wastewater:** incorporating clinical signals; simple rules with statistical guarantees; more high-quality data

Thank you.

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www.github.com/spertus/multivariate-change-point-detection/

References

Chu, C.-S. J., Stinchcombe, M., and White, H. (1996). Monitoring structural change. *Econometrica: Journal of the Econometric Society*, pages 1045–1065.

Cover, T. M. (1991). Universal portfolios. *Mathematical Finance*, 1:1–29.

Feng, J., Phillips, R., Malenica, I., Bishara, A., Hubbard, A., Celi, L., and Pirracchio, R. (2022). Clinical artificial intelligence quality improvement: towards continual monitoring and updating of ai algorithms in healthcare. 5.

Grünwald, P., de Heide, R., and Koolen, W. (2024). Safe Testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*.

Kishore, S. P. and Kocher, R. (2026). The hypertension control paradox — why is america stuck? *New England Journal of Medicine*, 394(5):417–420.

Lyu, Y. and Nkurunziza, S. (2024). Inference in generalized exponential o-u processes with change-point. *Statistical inference for stochastic processes*, pages 63–102.

Mei, Y. (2006). Sequential change-point detection when unknown parameters are present in the pre-change distribution. *Annals of Applied Statistics*, 34:92–122.

Orabona, F. and Jun, K.-S. (2022). Tight concentrations and confidence sequences from the regret of universal portfolio.

Ramdas, A., Grünwald, P., Vovk, V., and Shafer, G. (2023). Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576 – 601.

Shewart, W. A. (1931). *Economic control of manufactured products*. Van Nostrand Reinhold.

Shin, J., Ramdas, A., and Rinaldo, A. (2024). E-detectors: a nonparametric framework for sequential change detection. *The New England Journal of Statistics in Data Science*, 2.

Shiryaev, A. N. (1963). On optimum methods in quickest detection problems. *Theory of Probability & Its Applications*, 8(1):22–46.

Spertus, J. V. (2024). Cobra: Comparison-optimal betting for risk-limiting audits. In Essex, A., Matsuo, S., Kulyk, O., Gudgeon, L., Klages-Mundt, A., Perez, D., Werner, S., Bracciali, A., and Goodell, G., editors, *Financial Cryptography and Data Security. FC 2023 International Workshops*, pages 95–109, Cham. Springer Nature Switzerland.

Spertus, J. V., Glazer, A., and Stark, P. B. (2025). Dice, butnbsp;don't slice: Optimizing thenbsp;efficiency ofnbsp;oneaudit. In *Electronic Voting: 10th International Joint Conference, E-Vote-ID 2025, Nancy, France, October 1–3, 2025, Proceedings*, page 175–190, Berlin, Heidelberg. Springer-Verlag.

Spertus, J. V., Sridhar, M., and Stark, P. B. (2024). Sequential stratified inference for the mean. *arXiv preprint arXiv:2409.06680*.

Vovk, V. (2021). Testing randomness online. *Statistical science*, 36(4):595–611.

Vovk, V. and Wang, R. (2021). E-values: Calibration, combination and applications. *The Annals of Statistics*, 49(3):1736–1754.

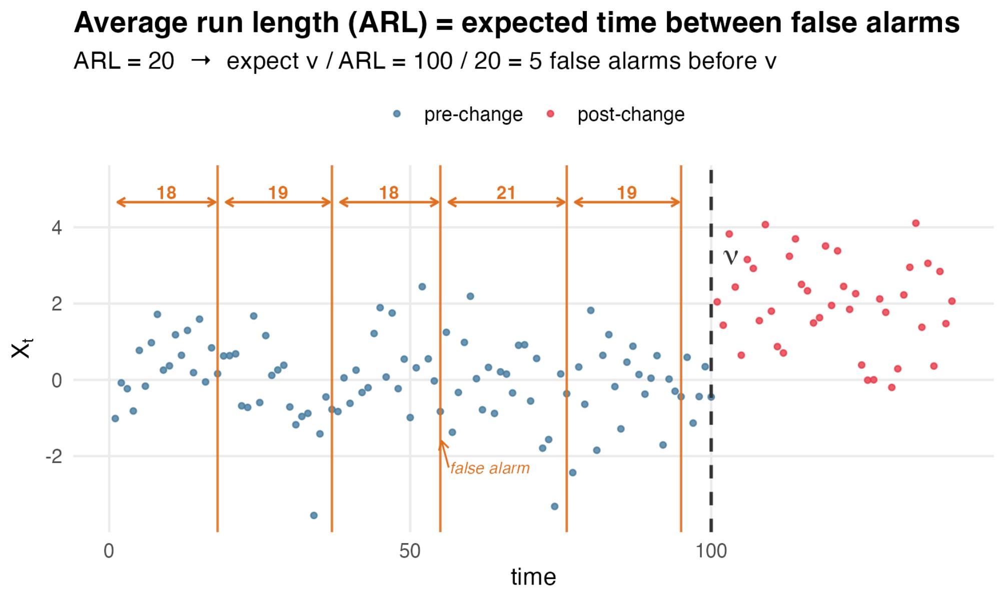
Waudby-Smith, I. and Ramdas, A. (2024). Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 86.

Waudby-Smith, I., Sandoval, R., and Jordan, M. I. (2025). Universal log-optimality for general classes of e-processes and sequential hypothesis tests.

Willsky, A. and H., J. (1976). A generalized likelihood ratio approach to the detection and estimation of jumps in linear systems. *IEEE Transactions on Automatic Control*, 21:108–112.

Xie, Y. and Siegmund, D. (2013). Sequential multi-sensor change-point detection. *The Annals of Statistics*, 41:670–692.

Appendix: ARL Intuition



Appendix: bounded model and AGRAPA bets

Model assumptions: $\boldsymbol{X}_t \in [0,1]^K$, $\mathbb{E}_P[X_{tk} \mid \mathcal{F}_{t-1}] \leq \eta_k$.

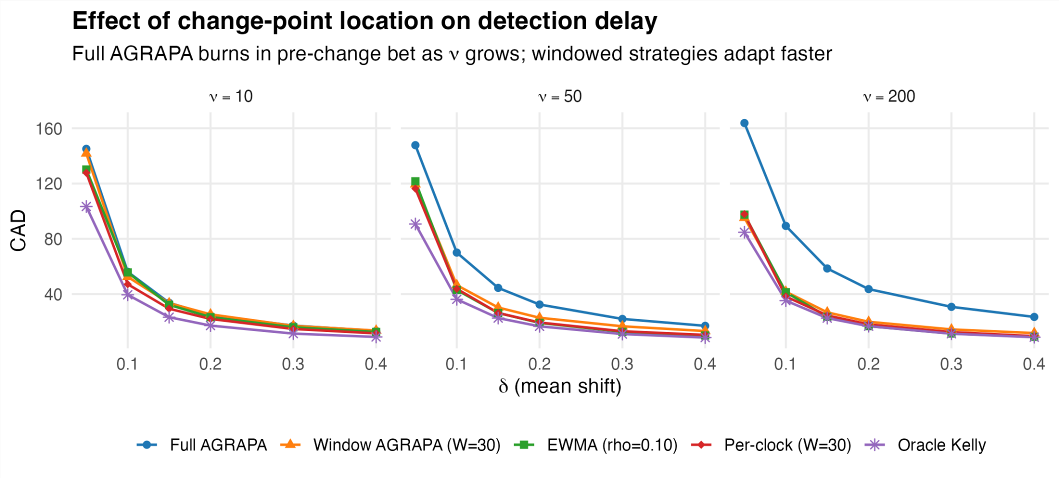
Marginal betting tsm.

$$M_t^k = \prod_{i=1}^{t} \left[1 + \lambda_{ik}(X_{ik} - \eta_k)\right], \qquad \lambda_{ik} \in [0, 1/\eta_k] \text{ predictable.}$$

AGRAPA (Waudby-Smith and Ramdas, 2024)

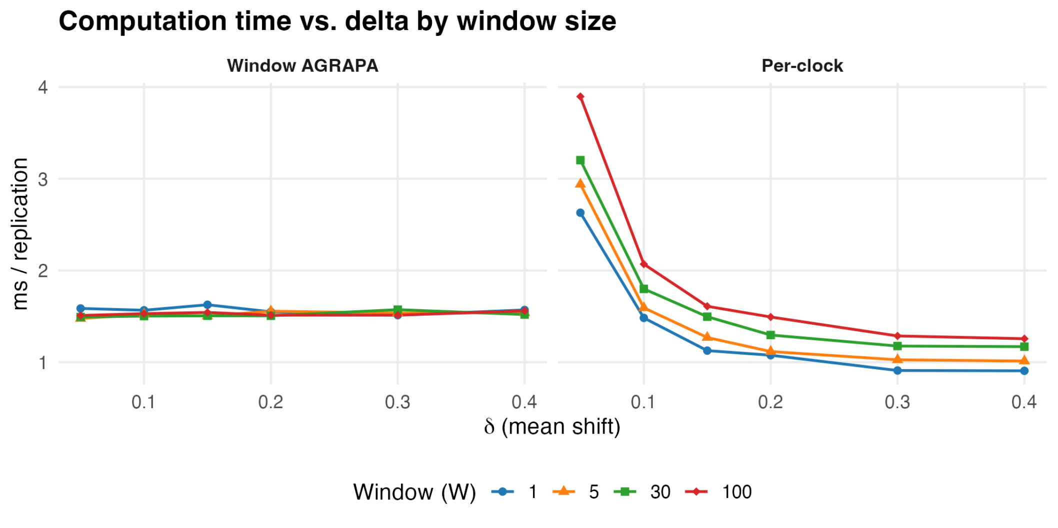
$$\lambda_{tk} = 0 \vee \frac{2(\hat{\mu}_{k(t-1)} - \eta_k)}{\hat{\sigma}^2_{k(t-1)} + (\hat{\mu}_{k(t-1)} - \eta_k)^2} \wedge \frac{c}{\eta_k}.$$

Appendix: CAD for bet windowing strategies over ν



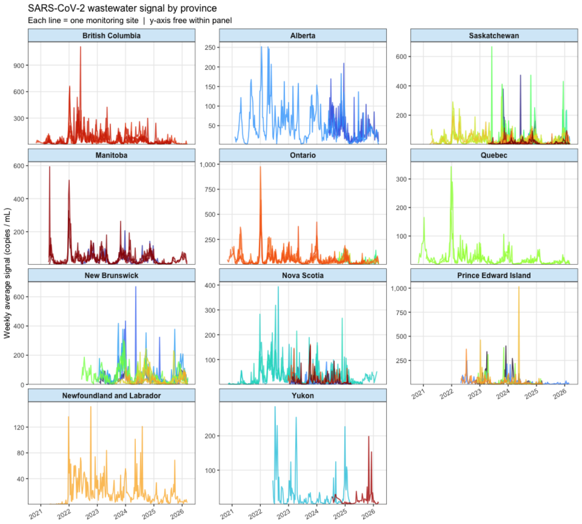
Conditional average delay of bounded TSMs + S-R procedure using different AGRAPA strategies

Appendix: Computation time for bet windowing strategies



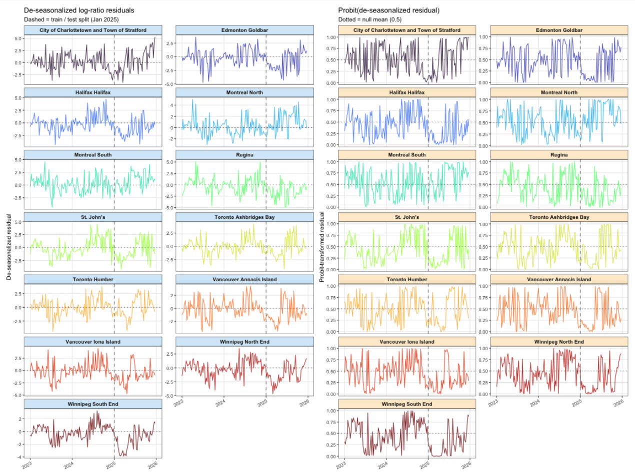
Computation time for bounded TSMs + S-R procedure using different AGRAPA strategies

Appendix: Raw NWMP Data



Raw covid signals over time at selected wastewater sites

Appendix: data transformations



Viral signals at 13 selected sites after normalizing by RSV and taking log ratio (left panel) and after applying probit transform (right panel); dashed line indicates training/test split time ('2025-01-01')

Appendix: joint detectors via meta-analysis of TSMs

Goal: detector fires if *any* stream changes

Let (M_{tk}) be any marginal TSM for stream k, with increments $\Lambda_{tk} := M_{tk}/M_{(t-1)k}$.

Three ways to combine:

- **Product** $M_t^p = \prod_k M_{tk}$ (valid *iff* streams are independent)
- **Average** $M_t^a = \sum_k w_k M_{tk}$ with $\sum w_k = 1$ (valid under *arbitrary* dependence)
- **Universal portfolio** $M_t^u = \int_{\mathcal{S}^K} \prod_{i=1}^t \sum_{k=1}^K u_k \Lambda_{ik} \, d\boldsymbol{u}$ (always valid, better than average)

Asymptotically, $M_t^u \geq \max_k M_{tk}$ (Cover, 1991), but M_t^u is hard to compute

Analogue: intersection testing (meta-analysis) with E-values (Vovk and Wang, 2021)

Appendix: localization

Goal: K change points $\{\nu_k\}_{k=1}^K \implies K$ detectors

Require $\boldsymbol{D} = [D_1, \ldots, D_K]$ to control the *global* ARL:

$$\mathrm{ARL}(\boldsymbol{D}) := \sup_P \mathbb{E}_{P,\infty}\left[\min_k D_k\right]$$

Testing analogue: multiple testing / family-wise error rate.

Union bound: set each detector's threshold to K/α

This is the Bonferonni correction, which is crude...

Appendix: spending allowance

A spending allowance (γ_t) has $\sum_k \gamma_{tk} = 1$ at *each* time.

Theorem: Thresholding each $R^{\gamma}_{tk} := \Lambda_{tk}(R^{\gamma}_{(t-1)k} + \gamma_{tk})$ controls the global ARL

(γ_t) may be predictable, adapting to past data + side information

Examples:

- $\gamma_{tk} := 1/K \implies$ Bonferonni
- Weighted $\gamma_{tk} \implies$ prioritize more important streams
- Bayesian posterior on $\gamma_{tk} \implies$ model on alarms
- Graph-adaptive $\gamma_{tk} \implies$ nearby sites more likely to alarm together (clusters)

Appendix: full ARL theorem

Theorem (S–R is ARL-valid).
Given any $\textsc{tsm}$ (M_t) under $\mathcal{P}$ with increments $\Lambda_t = M_t/M_{t-1}$, $R_t := \Lambda_t(R_{t-1}+1)$ yields $D_{\text{S-R}} := \inf\{t : R_t > 1/\alpha\}$ with $\sup_{P\in\mathcal{P}} \mathbb{E}_{P,\infty}[D_{\text{S-R}}] \geq 1/\alpha$.

Sketch. $R_t - t$ is a P-supermartingale; apply Doob's optional stopping theorem at $D_{\text{S-R}}$ and use $R_{D_{\text{S-R}}} \geq 1/\alpha$.

Appendix: full PFA theorem

Theorem (scheduled S–R is PFA-valid).
With spending schedule (π_t), $\sum_t \pi_t = 1$, and $R_t^\pi := \Lambda_t(R_{t-1}^\pi + \pi_t)$, the detector $D^\pi := \inf\{t : R_t^\pi > 1/\alpha\}$ satisfies $\sup_{P\in\mathcal{P}} \mathbb{P}_{P,\infty}(D^\pi < \infty) \leq \alpha$.

Sketch. Telescoping gives $R_t^\pi = \sum_i \pi_i M_t^{(i)}$, a convex combination of delayed $\textsc{tsm}$s—hence itself a $\textsc{tsm}$. Apply Ville.

Appendix: spending schedules for PFA control

A spending schedule (π_t) is a pmf on $\mathbb{N}$: $\sum_t \pi_t = 1$. Run a *scheduled* S–R:

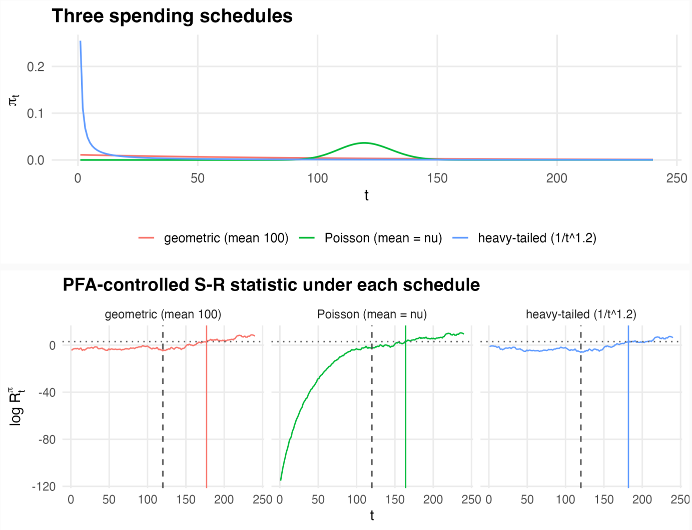
$$\boxed{R_t^\pi := \Lambda_t \cdot (R_{t-1}^\pi + \pi_t)} \qquad D^\pi := \inf\{t : R_t^\pi > 1/\alpha\}.$$

Theorem (this work). $\mathbb{P}_{P,\infty}(D^\pi < \infty) \le \alpha$.

Investment view. Instead of a fresh \$1 each step, spend only π_t at time t; total lifetime budget capped at \$1.

(π_t) is *like* a prior on ν, but validity does *not* depend on the prior being correct—only efficiency does.

Appendix: spending schedules and their detectors



Spending schedules and PFA-controlled S-R statistic under each. Concentrated schedules are efficient when ν sits in their bulk; wider schedules hedge over a range of ν.

Appendix: What is the "state of control" in NWMP?

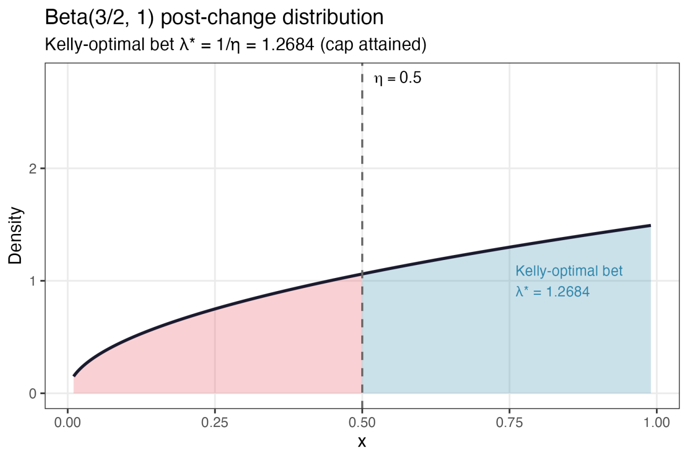
Even with perfect data, the target is ambiguous. A working definition:

Operational state of control: Covid signal moves *in line with* a reference virus (RSV, pMMoV) and with the historical seasonal contour observed in 2023–2025.

Concretely: Normalize weekly covid signal by reference virus, de-seasonalize, and treat residuals as the monitored stream. Detection fires when covid is *unseasonably high* relative to its long-run movement with the reference.

Caveat: Different stakeholders may consider different states of control important (e.g. absolute thresholds, residual correlation with leading hospitalizations).

Appendix: Beta Q and Kelly bet



The alternative distribution Q chosen for the wastewater example + induced Kelly optimal bet λ^*